Slopes in Surface Mine

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1.0 INTRODUCTION

Slopes in Surface mine can be divided into two groups first being the 'High Wall Slopes' and the second being 'Dump Slopes'. High wall slopes are often called bank slopes and constitute face or front slopes and side slopes. Such slopes are formed by excavation of in situ rock mass or soil. On the other hand dump slopes are formed by excavated overburden, waste rock or mill tailings etc. Judicious establishment of geometric limits of pit is closely related to the profitability of any mine. On one hand stable slopes are essential for uninterrupted production of ore or coal on the other hand steep slopes are desirable to reduce stripping ratio consequently to extend the life and to improve the economics of an open pit. These two conflicting requirements of technical and economic compromise thus pose a challenge for the mining engineer to design a most efficient slope for the mine with emphasis on its stability. The primary objective of the mining engineer with respect to slope stability thus boils down to the following:

i. Finding the endangered area
ii. Investigation of failure mechanism
iii. Sensitivity to failure (parametric analysis)
iv. Design of optimal slope with respect to safety
v. Reliability and economics
vi. Design possible remedial measures.

2.0 CAUSES OF SLOPE FAILURES

Failure occurs when the loads or stresses acting on the rock material (intact or discontinuous) exceed the strength (compressive or tensile) of the rock. The considerations which govern the failure or stability of a slope are described as follows:

- **Geological structure**: Geological structure includes both rock type and discontinuities which together form the rock mass. In a rock mass, discontinuities exist virtually in all scales. The most probable failure surfaces are usually along the foliation joint of discontinuities. Other geological discontinuities include joints, which vary in size, and presence of faults. Faults are distinguished from joints as it have undergone shear movements.

- **Rock stresses and ground water conditions**: Knowledge of stress state in a slope is essential in order to understand the mechanics of slope behavior. The stresses acting upon a structure in comparison to the strength of structure govern the stability. The virgin stresses in rock mass are usually compressive in nature and are due to gravitational and tectonic considerations. Virgin vertical stress at a point is due to the weight of overlying rock mass, whereas, virgin horizontal stresses are due to tectonic normally present.

The stress state in a slope is also dependent on groundwater conditions. Groundwater is defined as water below the water table, which is also known as zone of saturation. The presence of ground water reduces effective stress in the rock mass, thus reducing the shear strength along the probable failure surface. Groundwater acts as an additional driving force causing failure. Erosion brought due to flowing water could also result in reduced strength

- **Strength of discontinuities and intact rock**
- **Slope geometry**
- **Vibrations from blasting or seismic events**
- **Climatic conditions**
- **Time**

These factors independently or in combination poses complex situation in design of slopes stability. These factors also determine the mode of failure of a slope. It is necessary to understand the modes of failure and governing mechanism prior to design.

3.0 MODES OF SLOPE FAILURE

Gravitational and seepage forces are main cause of instability in slopes. The slope may be appropriated into soil if it is made of highly weathered material or highly jointed rock mass. Such slopes have to be analyzed using soil mechanisms techniques with the failure mode being circular. A distinction can also be made between the mechanisms that occur in hard and in softer rock types. Most of the rock slopes tend to fail along the pre-existing planes of weakness such as joints, bedding planes or faults. In rock slopes, type of failure is primarily controlled by the orientation and spacing of discontinuities within the rock mass as well as relative orientation of excavation and angle of inclination of the slope. Modes of failure in rock mass are either circular, planar, wedge or toppling. The classification for the slide mechanism according to Franklin and Dusseault is explained as follows.

Seven failure mechanisms can be associated with the sliding failure mode in rock slopes. While other failure mechanisms are conceptually possible, the seven mechanisms illustrated are representative of those mechanisms most likely to occur.

a. **Single block/single sliding plane**: A single block with potential for sliding along a single plane (Figure 1a) represents the simplest sliding mechanism. The mechanism is kinematically possible in cases where strike direction of at least one joint set is approximately parallel to the slope strike and dips toward the slope excavation. Failure is impending if the joint plane intersects the slope plane and the joint dips at an angle greater than the angle of internal friction (ϕ) of the joint surface.

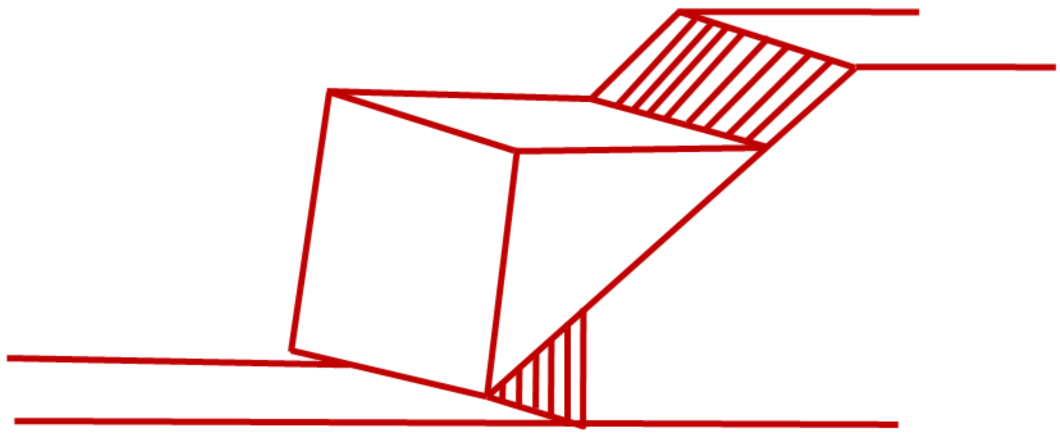


Figure 1a: Single block single sliding plane of failure

b. **Single block/stepped sliding planes**: Single block sliding along stepped planes (Figure 1b) is possible in cases where strike of a series of closely spaced parallel joints is approximately parallel to the excavation slope strike and dips towards the excavation slope. The parallel joints may or may not be continuous. However, at least one joint plane must intersect the slope plane. In the case of continuous parallel joints, a second set of joints is necessary for causing failure. This second joint set must also strike more or less parallel to the slope and the magnitude and direction of the joint dip angle must be such that the joint plane does not intersect the slope plane.

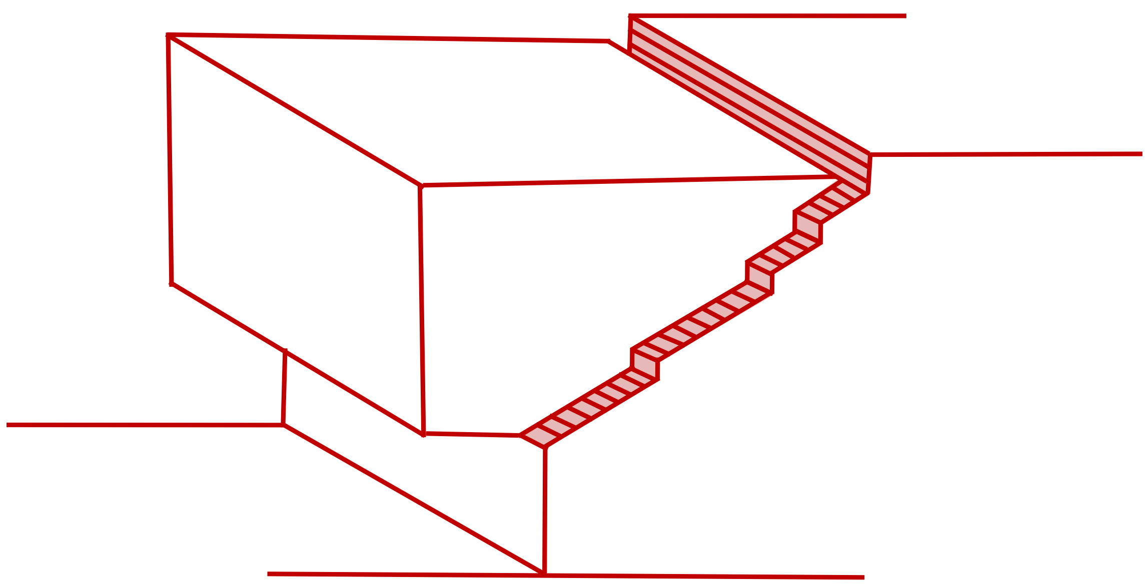


Figure 1b: Single block/stepped sliding planes

c. **Multiple blocks/multiple sliding planes**: Multiple blocks, sliding along multiple planes (Figure 1c) is the most complicated planar type of sliding. The mechanism is associated with two or more joint sets that strike approximately parallel to the slope strike and dip in the direction of the excavation slope. At least one of the joint planes must intersect the excavated slope plane. For a failure to occur, the dip angle of the joint defining the base of the upper most block must be greater than the frictional angle of the joint surface. Furthermore, additional joints must be present which also strike approximately parallel to the strike of the excavated slope. These additional joints must either dip in a near vertical direction or dip steeply away from the slope plane.

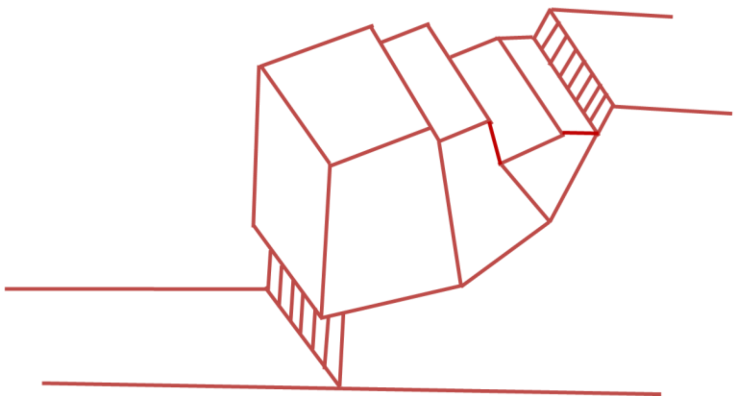


Figure 1c: Multiple blocks/multiple sliding planes

d. **Single wedge/two intersecting planes**: Single wedge sliding (Figure 1d) can occur in rock masses with two or more sets of discontinuities whose lines of intersection are approximately perpendicular to the strike of the slope and dip towards the plane of the slope. In addition, this mode of failure requires that the dip angle of at least one joint-set is greater than the frictional angle of the joint surfaces and that the line of joint intersection intersects the plane of the slope.

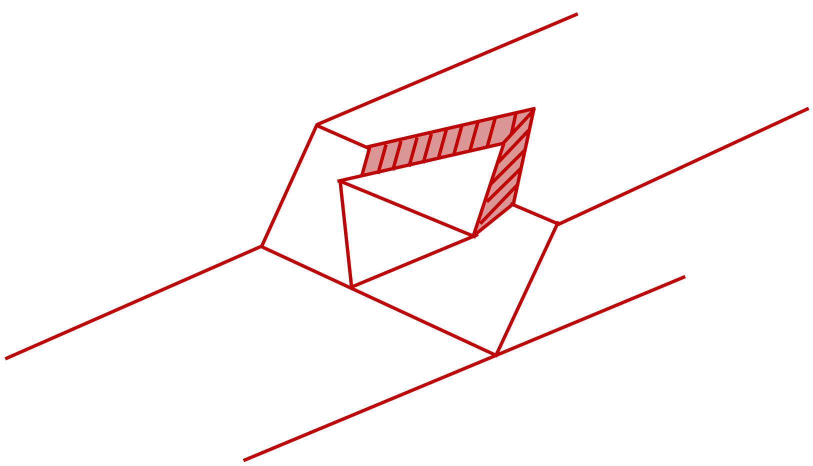


Figure1d: Single wedge/two intersecting planes

e. **Single wedge/multiple intersecting planes**: The conditions for sliding of a single wedge formed by the intersections of at least two discontinuity sets with closely spaced joints are essentially the same as discussed in point (d) above (Figure 1e).

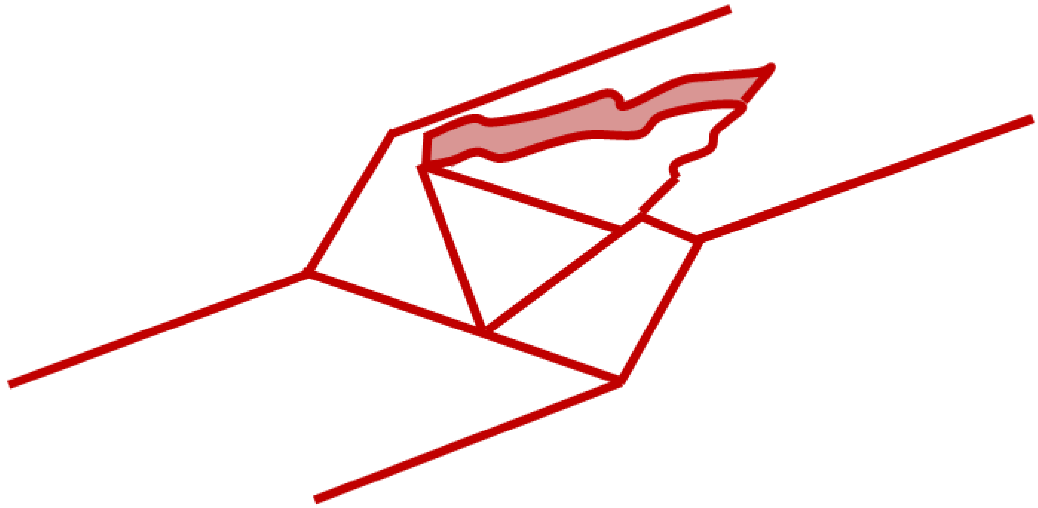


Figure 1e: Single wedge/multiple intersecting planes

f. **Multiple wedges/multiple intersecting planes**: Multiple wedges can be formed by the intersection of four or more sets of discontinuities (Figure 1 f). Although conceptually possible, the sliding failure of a multiple wedge system rarely occurs because of the potential for kinematic constraint.

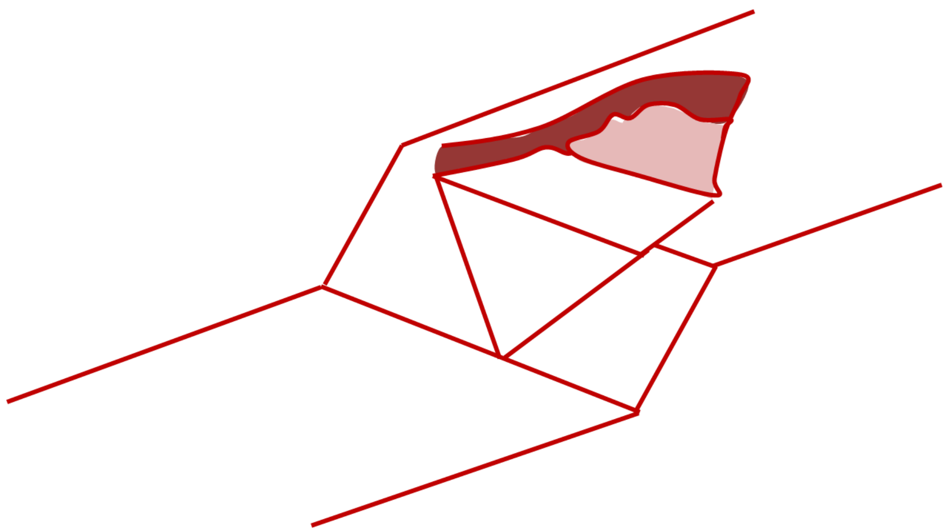


Figure 1f: Multiple wedges/multiple intersecting planes

g. **Single block/circular slip path**: Single block sliding failures along circular slip paths are commonly associated with soil slopes. However, circular slip failures may occur in highly weathered or decomposed rock masses, highly fractured rock masses, or in weak rock such as clayey shales and poorly cemented sandstones.

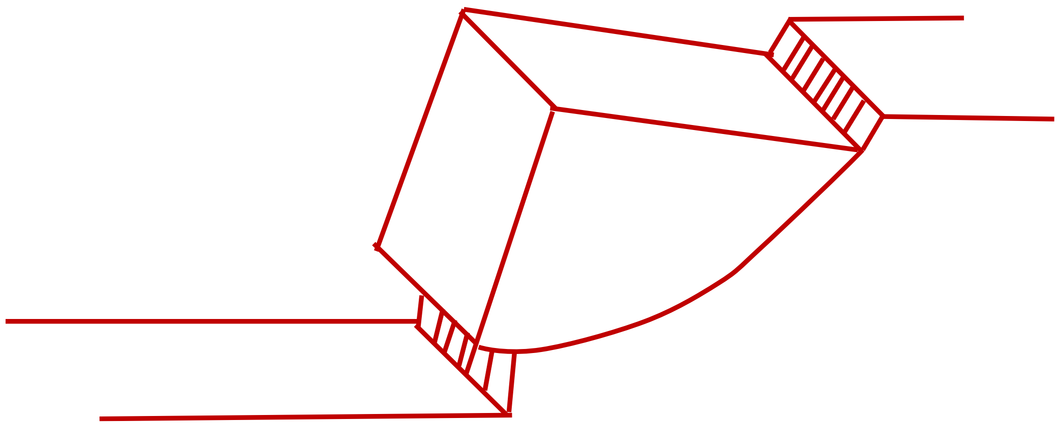


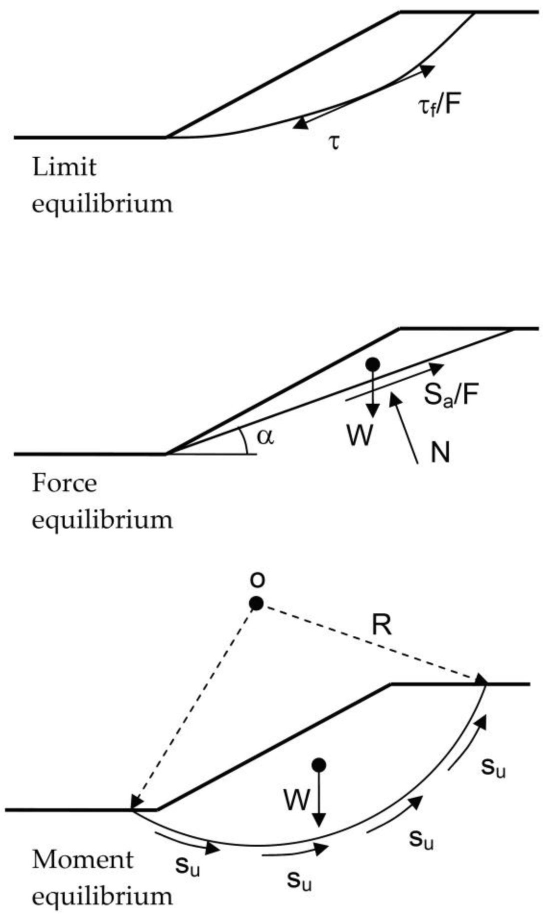
Figure 1g: Single block/circular slip path

4.0 METHODS OF SLOPE STABILITY ANALYSIS

Analysis of stability involves the identification of the mode of failure, estimating the shearing resistance along the plane of sliding and evaluating the degree of safety using a suitable criterion. The commonly used methods adopted for slope stability analysis are Limit Equilibrium Methods, Numerical Methods, Physical Modeling, and Neural Networks. Among all, limit equilibrium method is most widely used. It is considered that failure occurs along an assumed or a known failure surface. The shear strength required to maintain a condition of limiting equilibrium is compared with the available shear strength of slope material, giving the average factor of safety along the failure surface. The limit equilibrium method considers the problem as two dimensional, by assuming conditions of plane strain. Numerical methods such as Finite, Boundary and Distinct element programs can be used for slope stability analysis. In the first two methods, the rock mass is modeled as a continuum. In the distinct element method, the slope can be modeled as a reality model. If it is attempted to model the slope as a continuum, the problems inherent to scale effects and transforming discontinuity data into a continuum deformation model have to be solved first. In case the problem is modeled as a reality model, as in a distinct element method, the number of blocks and consequently the quantum of calculation time and computer memory increase rapidly for larger models. In the finite element method, the stresses and strains in entire slope are evaluated by elastic analysis and stability is checked not only by predicting factor of safety against shear failure, but also by undesirable displacements.

4.1 Limit Equilibrium Method of Slope Stability

All limit equilibrium methods utilise the Mohr-Coulomb expression to determine the shear strength (τ) along the sliding surface. A state of limit equilibrium exists when the mobilised shear stress (τ) is expressed as a fraction of the shear strength. At the moment of failure, the shear strength is fully mobilised along the failure surface when the critical state conditions are reached. The Factor of Safety (FoS) can be defined in three ways: Limit equilibrium, force equilibrium and moment equilibrium as shown in Figure 2 below.



Limit equilibrium:

$$F = \frac{s_u}{\tau} \qquad \text{(Total stress)}$$

$$F = \frac{c' + \sigma' \tan\phi'}{\tau} \qquad \text{(Effective stress)}$$

Force equilibrium:

$$F = \frac{\text{Sum of resisting forces}}{\text{Sum of driving forces}}$$

$$F = \frac{S_a}{W \sin\alpha} = \frac{\mathrm{cL+Ntan}\phi}{\mathrm{Wsin}\alpha}$$

where,

L = total length of the sliding plane

Moment equilibrium:

$$F = \frac{\text{Sum of resisting moments}}{\text{Sum of driving moments}}$$

$$F = \frac{R \int_0^L s_u \, dl}{W.x}$$

Figure 2: Definition of Factor of Safety by three ways

4.2 The Finite Element Method

FEM analysis constitutes three steps mainly, domain discretization, local approximation and assemblage and solution of global matrix equation. This method involves the representation of continuum as an assembly of elements which are connected at discrete points called nodes. The problem domain is divided into discrete elements of various shapes, e.g. triangles and quadrilaterals in two-dimension cases and tetrahedrons and bricks in three dimensions. All forces are assumed to be transmitted through the body by the forces that are set up at the nodes. Expressions for these nodal forces, which are essentially equivalent to forces acting between elements, are required to be established. Continuum problem is analyzed in terms of sets of nodal forces and displacements for the problem domain.

The displacement components within the finite elements are expressed in terms of nodal displacements. Derivation of these displacements describes strain in the element.

The stiffness of the medium to this induced strain determines stress in the element. Total stress within an element can be found out by superimposition of initial and induced stresses. The matrix of each element describes the response characteristics of the elements. These coefficient matrices are based on minimization of total potential energy. The elemental stiffness matrices are assembled to give the global stiffness matrix which is related to global force and displacement. As the number of elements in a problem domain tends to infinity, this is equivalent to solving differential equation.

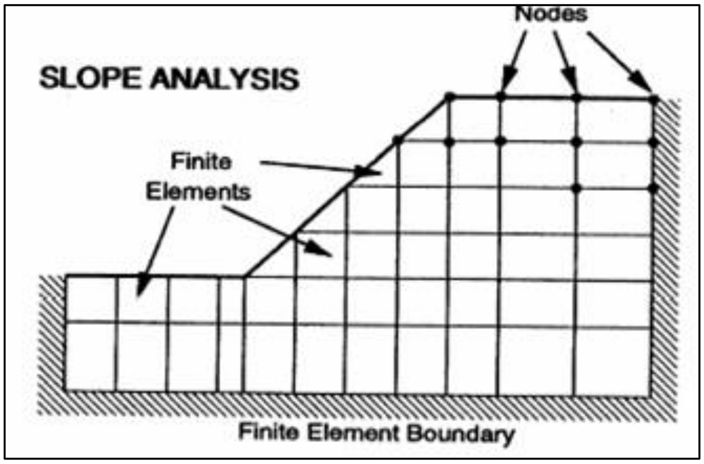


Figure 3: Finite element modelling

FEM suffers limitations when simulating fracture problems mainly due to the limitation of small element size, continuous re-meshing with fracture growth, conformable fracture path and element edges. Disadvantage of this method is that considerable time is required in preparing input data for a typical problem. This is particularly crucial in 3D problems. FEM is computationally expensive. A large number of simultaneous equations must be solved to obtain a solution. If the problem is non-linear, the computation time increases enormously because the sets of simulataneous equations must be solved a number of times.

Inspite of all the above disadvantages, FEM is widely used due to its generality and flexibility to handle material heterogeneity, non-linearity and boundary conditions, with many well developed commercial codes with large capacity in terms of computing power, material complexity and user friendliness. It is one of several well-developed techniques that can provide useful information for engineering surface and underground excavations in rock.

Some of the commercial packages available are as follows:

- ANSYS (ANSYS Inc.) – http://www.ansys.com
- Phase2 (Rock Science) – http://www.rockscience.com
- PLAXIS (PLAXIS BV) – http://www.plaxis.nl

4.3 Finite difference method (FDM)

This is one of the oldest numerical techniques used for the solution of sets of differential equations, given initial values and/or boundary values. The difference equations for a triangle are derived from the generalized form of Gauss' divergence theorem. Differential equations are solved by dividing the domain into connected series of discrete points called nodes. These nodes are the sampling points for the solution and are linked using finite difference operators to the governing equations. It is not necessary to combine the element matrices into a large global stiffness matrix as in the FE model. Instead, the FD method regenerates finite difference equations at each step. Derivatives of governing equations are replaced directly by algebraic expressions written in terms of field variables, e.g. stress or displacement, at discrete points in space (nodes).

The Finite Difference method allows one to follow a complicated loading path and highly non-linear behaviour without requiring the complex iterative procedure of a standard implicit code. Finite difference method can be used to discretize both time and space. It also provides easy error estimation techniques. It is particularly suitable for large, non-linear problem which may involve collapse or progressive failure.

Finite difference method is difficult to use for irregular shape domain because the fine meshing required near the singularity cannot be easily reduced for the rest of the domain. The conventional FD method with regular grid systems does suffer from shortcomings, most of all in its inflexibility in dealing with fractures, complex boundary conditions and material heterogeneity. This makes the standard FD method generally unsuitable for modelling practical rock mechanics problems. However, with the use of irregular meshes (triangular grid or Voronoi grid systems), which leads to Control Volume or Finite Volume techniques, significant progress has been made. FLAC (Itasca, USA – http://www.itascacg.com) is the well-known computer code for stress analysis for engineering problems using FVM/FDM approach. Explicit representation of fractures is not easy in FD method as they require continuity of the functions between the neighboring grid points. In addition, it is not possible to have special fracture elements as in FE method. However, this is the most popular numerical methods in rock engineering with applications covering from all aspects of rock mechanics, e.g., slope stability, underground openings, coupled hydro-mechanical etc.

4.4 Boundary element method (BEM)

Rock mass is predominantly very large and for practical purposes can be assumed to be of infinite extent. Because of its volume discretization the Finite element is not very well suited for problems with a low ratio of boundary surface to volume since a large number of elements are required to model the response of the domain. Boundary element method is particularly attractive for such analyses in rock mechanics where the surface of the excavation has to be discretised. The amount of input data required to describe a problem is greatly reduced and the influence of infinite rock mass is automatically considered in the rock mass.

The Boundary Element method requires the discretisation of the domain and, if necessary the boundaries between the regions with different properties. For two-dimensional situations line elements at the boundary represent the problem, while for three-dimensional problems, surface elements are required. Thus, the dimensionality of the problem is reduced by one. This is particularly attractive as the amount of data required to describe the problem is greatly reduced as compared to FEM. Due to the BEMs advantage in reducing model dimensions, 3D application using the displacement discontinuity method for stress analysis has become efficient.

Although BEM method has been used in complex rock mechanics problems involving non-linear constitutive equations and number of materials, this method of analysis is particularly efficient in homogeneous, linear elastic problems in three dimensions. Advantage of this method is reduced in complex non-linear material laws with sets of materials because the surface needs be discretised wherever there is change of material properties, and hence the preparation of input data becomes more complicated. It is necessary to perform integrations over the volume of the rock mass for such an inhomogeneous rock mass or one with nonlinear behaviour and so some of the advantage over other methods are lost. Matrices of the equations in this method are not symmetric and banded as in FEM. Though the number of equations to be solved is less, the computation time is not reduced by the same amount.

4.5 Dis-Continuum Approach

Rock joints and discontinuities in a rockmass play a key role in the response of a rock slope, tunnel or excavation, i.e. joints can create loose blocks near the surface or tunnel profile and cause local instability; joints weaken the rock and enlarge the displacement zone caused by excavation; joints change the water flow system in the vicinity of the excavation. The use of discontinuum modelling has been gaining progressive attention in slope engineering and tunnel engineering mainly through the use of the UDEC and 3DEC codes (Itasca, USA), for 2D and 3D discontinuum modelling respectively.

In the distinct element method, the rock mass is represented as an assemblage of discrete blocks which may be considered either "not deformable" or "deformable". Joints and discontinuities are viewed as interfaces between distinct bodies. The important features of this method which make it appropriate in order to capture the important mechanisms characterising a discontinuous medium are

(i) the method allows finite displacements detachment (as shown in Figure 4)

(ii) the method recognizes new contacts automatically as the calculation progresses.

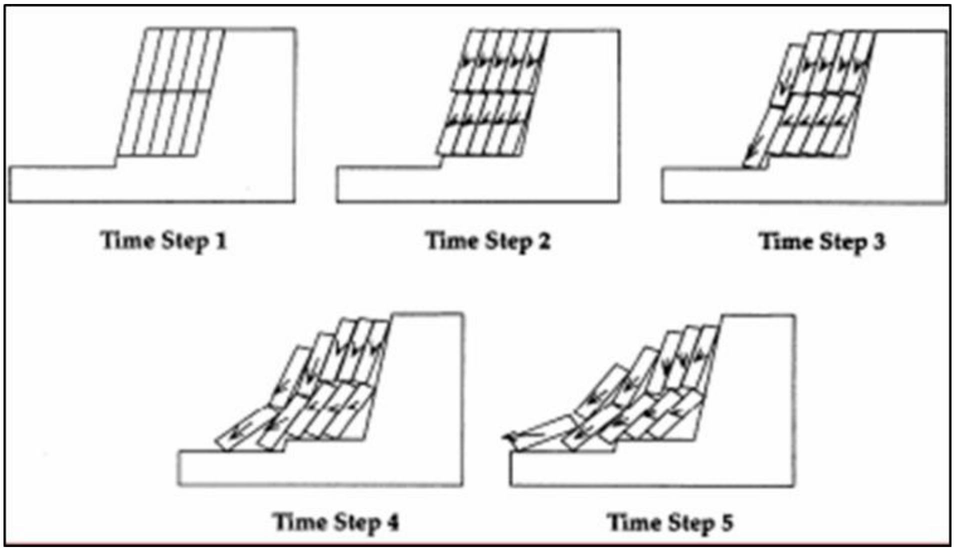


Figure 4. A discontinuum model

In order to apply the distinct element method to the solution of problems, there are two crucial issues which include the joint geometry data and the material properties assigned to the joints. The distinct element method is the popular method in the dis-continuum Approach.

4.6 Distinct Element Method (DEM)

DEM is one of the most rapidly developing areas of computational mechanics with a broad variety of applications in rock mechanics. UDEC and 3DEC are the most popular computer codes used to perform static as well as dynamic analysis.

This method treats domain as a discontinuum rather than a continuum in contrast to FEM and BEM. In DEM, rock mass is treated as an assemblage of rigid or deformable discrete blocks/particles. The contact displacements at the interfaces of a stressed assembly of blocks are identified and continuously updated throughout the deformation process and represented by appropriate constitutive models. The elements interact with one another through the forces developed at contact points. The equations of equilibrium are repeatedly solved until the laws of contacts and boundary conditions are satisfied.

Discrete element technique method is capable of analyzing multiple interacting deformable continuous, discontinuous or fracturing bodies undergoing large displacements and rotations. Dynamic equilibrium equation is solved for each body subjected to boundary interaction forces. There is no restriction on where one element may make contact with another, and nodes may interact with nodes or nodes with element faces. Forces generated between the contacting elements can be made to obey various interacting laws depending upon the physical nature of simulation. For example, interaction relationships for rock joints may include cohesion, dilation, damage to asperities, and stress dependent friction. Since elements rapidly change neighbouring elements upon fracturing or motion, automatic algorithm is used to compute connectivity or interaction of element to element.

5.0 SLOPE STABILISATION AND PROTECTION MEASURES

5.1 Drainage and Reduction of pore water Pressure

Ground water in rock slopes is often the main contributory cause of instability. Reduction in water pressures improves the stability. Water pressure can be controlled by limiting surface infiltration and drilling horizontal drain holes or driving adits at the toe of the slope to create outlets for the water (Figure 5). The selection of the best method for the site will depend on the intensity of the rainfall, the permeability of the rock and the dimensions of the slope.

For sites that experience heavy rainfall and can rapidly saturate the slope and cause surface erosion, it is beneficial to construct drains both behind the crest and on benches on the face to stop the water. Such drains can be lined with masonry or concrete to prevent the collected water from infiltrating into the slope. The drains should be dimensioned such as to carry the expected peak design flows. The network of drains should be such that they are interconnected so that the water is discharged to the storm drain system or nearby water courses. Where the drains are on steep gradients, it is sometimes necessary to incorporate energy dissipation protrusions in the base of the drain to limit flow velocities. Periodic maintenance to keep the drains clear should be carried out to prevent rapid vegetation growth in climates with high rainfall.

Horizontal drain holes: A series of drain holes (inclined upwards at about 5◦) into the face of rock slopes are drilled. This serves as an effective means of reducing the water pressure in the slope. Proper precautions should be taken with respect to the alignment of the holes so that they intersect the discontinuities as most of the ground water is contained in discontinuities. There are no widely used formulae for calculation of the required spacing of drill holes, but as a guideline, holes are usually drilled on a spacing of about 3–10 m, to a depth of about one half to one-third of the slope height. It is always preferable that the holes are lined with perforated casing, with the perforations sized to minimize infiltration of fines that are washed from fracture infillings. Drain holes can be drilled to depths of several hundred meters, sometimes using drilling equipment that installs the perforated casing as the drill advances to prevent caving. Also, it is common to drill a fan of holes from a single set up to minimize drill moves.

Drainage adits: For large slopes, it may not be possible to reduce the water pressure in the slope significantly using relatively small drain holes. In these circumstances, a drainage tunnel may be driven into the toe of the slide from which a series of drain holes are drilled up into the saturated

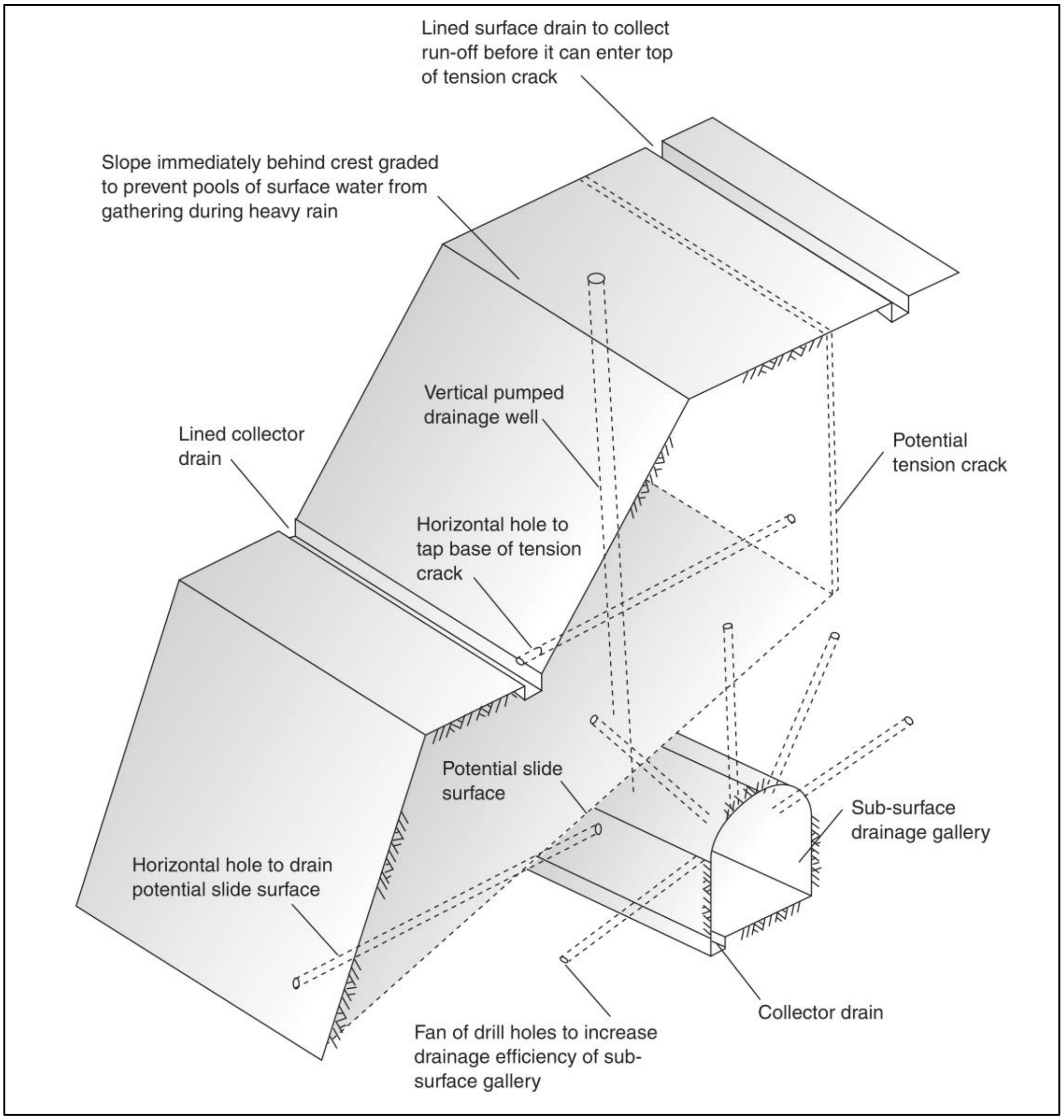


Figure 5: Drainage and reduction of pore water pressure for slope stabilization

5.2 Modification of Slope Geometry

Removal of rock is also one of the preferred method of stabilization of slope because such activity will eliminate the hazard, and no future maintenance is required. However, removal should only be used where it is certain that the new face will be stable, and there is no risk of undermining the upper part of the slope. Area 4 on Figure 6 is an example of where rock removal should be carried out with care. It would be safe to remove the outermost loose rock, provided that the fracturing was caused by blasting and only extended to a shallow depth. However, if the rock mass is deeply fractured, continued scaling will soon develop a cavity that will undermine the upper part of the slope. Removal of loose rock on the face of a slope is not effective where the rock is highly weathered or degradable, such as shale. In such cases, exposure of a new face will simply start a new cycle of weathering and instability. For this condition, most useful stabilization methods would be protection of the face with shotcrete and rock bolts, or a tied-back wall.

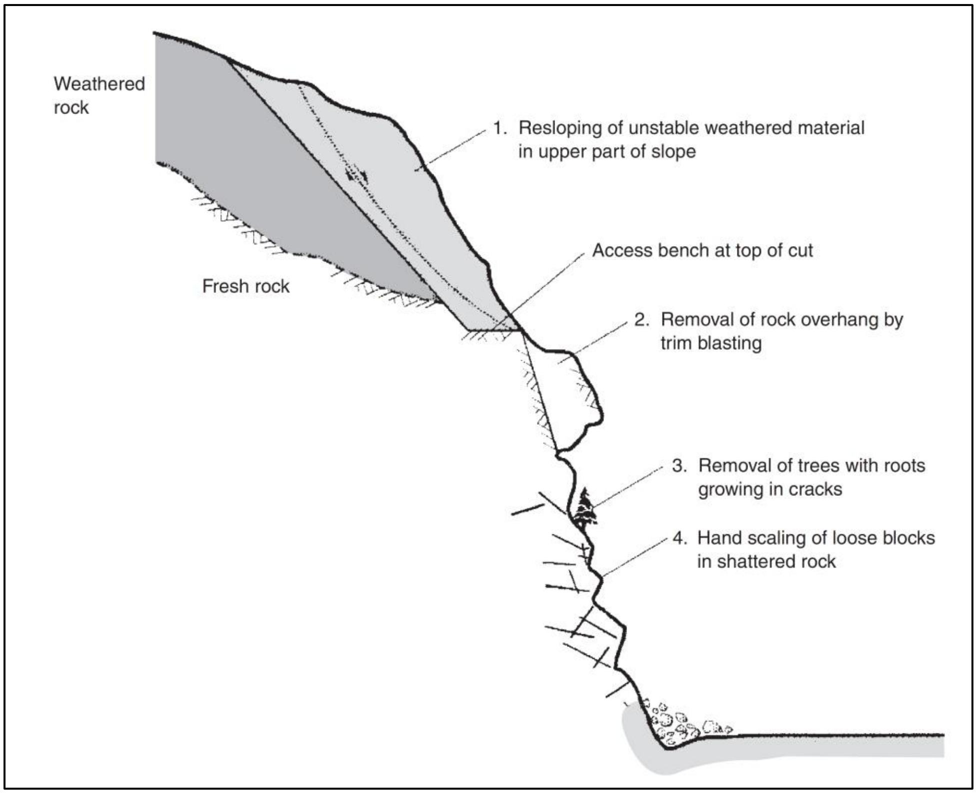


Figure 6: Slope stabilization by modification in slope geometry

Where overburden or weathered rock occurs in the upper portion of a cut, it is often necessary to cut this material at an angle flatter than the more competent rock below (Figure 6, item 1). Another condition that should be taken account of during design is weathering of the rock some years after construction, at which time re-sloping may be difficult to carry out. A bench can be left at the toe of the soil or weathered rock to provide a catchment area for minor slope failures and provide equipment access. Where a slide has developed, it may be necessary to unload the crest of the cut to reduce its height and diminish the driving force. Re-sloping and unloading is usually carried out by excavating equipment such as excavators and bulldozers. Consequently, the cut width must be designed to accommodate suitable excavating equipment on the slope with no danger of collapse of the weak material while equipment is working; this width would usually be at least 5m. Safety for equipment access precludes the excavation of "sliver" cuts in which the toe of the new cut coincides with that of the old cut.

Failure or weathering of a rock slope may form an overhang on the face (Figure 6, item 2), which could be a hazard if it were to fail. In these circumstances, removal of the overhang by trim blasting may be the most appropriate stabilization measure. Scaling involves the removal of loose rock, soil and vegetation on the face of a slope using hand tools such as scaling bars, shovels and chain saws.

5.3 Stabilization by rock reinforcement

A number of reinforcement techniques can be implemented to secure potentially loose rock on the face of a rock cut. The common feature of all these techniques is that they minimize relaxation and loosening of the rock mass that may take place as a result of excavation. Once relaxation has been allowed to take place, there is a loss of interlock between the blocks of rock and a significant decrease in the shear strength.

5.4 Slope Surface Protection by Vegetation

Vegetative techniques are most frequently used for stabilization of dumps. Generally, these methods are most successful when minor or shallow instability (such as raveling or erosion) is involved, as is usually the case for soil slopes or highly fractured rock slopes. The plantation of vegetation on steep soil slopes or loose-rock slopes is done by the construction of benches or stair-step terraces in the slope face. Chemical binders and soil stabilizers may also be used to temporarily stabilize the soil.

Geosynthetics can also be used for erosion control and to hold seeds in place, though their cost and effectiveness often limit their use. They require high labor inputs for installation in addition, some are not well adapted to fitting to rough surfaces. They must also be heavy enough or anchored in enough spots to prevent wind whipping. In order to make access easier for planting trees or shrubs, benches, berms, or furrows must be constructed in the slope face.

Slope Surface Protection by Gunite or Shotcrete

One common method utilizes pneumatically applied mortar and concrete (generally known as gunite or shotcrete) sprayed or pumped onto the slope face to seal the face and bind together small fragments on the face. This approach is used primarily to prevent weathering and spalling of a rock surface, as well as to knit together the surface of a slope. Generally, for rock slope stabilization, the material is applied in one 50- to 75-mm (2- to 3-in) layer. Shotcrete is applied by either wet or dry application. For dry mix shotcrete, additives and mortar are mixed on-site and pumped via compressed air to the nozzle, where the water is added. The wet mix is premixed at a central plant to specifications and then transported to the site in bulk.

One disadvantage of shotcrete is its low tensile strength. For this reason, welded wire mesh, anchored to the rock, is often used to reinforce the shotcrete. A problem with using wire mesh as reinforcement for shotcrete is the difficulty of molding the mesh to a rough surface. Where the surface is irregular, large gaps may develop between the mesh and the rock, making bonding of the shotcrete to the rock difficult. Additives can be added to either the wet or dry mix to provide additional strength and durability. Steel fibers, when added to the mix, increase the tensile strength of the shotcrete by providing numerous bonding surfaces within a small area. The fiber reinforcement also reduces the risk that shrinkage cracks will develop during curing. In many cases, the addition of fibers can replace wire mesh as reinforcement, thus reducing the overall cost.